

Probe Reliability is an Emergent Property of Trained Diffusion Models

Why training-free feature-caching selectors work—and why controlled-denoiser benchmarks mislead

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Abstract. Training-free diffusion accelerators (TAP, TeaCache, DiCache, ToCa) skip most of the denoising network and reconstruct its output from cached features, deciding *what to skip* by reading a cheap *probe*—the first-layer modulated input. These methods work well in practice, but *why* the probe is a good selector is unknown. We set out to improve one such method and instead found a more interesting fact: **the probe’s ability to select predictors is an emergent property of training, not of architecture.** Across diffusion transformers (DiTs) trained on **three real datasets** (MNIST, Fashion-MNIST, CIFAR-10), the rank correlation between the probe’s verdict and the true model-output error rises from 0.26–0.84 at initialization to a peak of 0.88–0.97 within a few hundred steps (tight bootstrap CIs), locked to the onset of denoising capability, then *relaxes* as capability keeps improving; MNIST and Fashion-MNIST trace nearly identical curves. The same arc reproduces—and *sharpens*—across a $3.7\times$ model scale-up (a 7.4M-parameter DiT-B peaks higher at 0.97 and relaxes deeper to 0.74), so the effect is not a small-model artifact. At convergence the cheapest first-layer probe is the most reliable of the first four layers on both grayscale datasets—directly justifying its use in practice. We stress-test everything with a pre-registered 55-test suite. We give a mechanistic account: the probe-error profile linearly predicts the true per-predictor error profile with $R^2=0.61$ (up from 0.42 early in training) and the selector picks the *truly best* predictor 82% of the time (chance 7%); its variance is low-dimensional (97% in two directions), yet a causal test shows selection still needs the full profile, and high-displacement tokens are assigned higher-order predictors. Two consequences follow. First, per-step optimality is the wrong objective: a per-step oracle is *worse* than probe-greedy (0.45 vs. 0.25 deviation), because locally-best predictors seed errors that compound. Second, the same method comparisons *flip sign* between controlled (analytic / random-feature) denoisers and a trained DiT—so these accelerators must be evaluated on trained networks with fidelity-to-base-model metrics. Our own attempted “improvements” to TAP win on controlled testbeds and lose on the trained DiT, illustrating the hazard.

All numbers are computed from experiments run on a single Apple M3 (MPS) and released with the paper. We report a small, real, trained DiT and do not claim FLUX-scale FID.

1 Introduction

A diffusion model generates by integrating a learned denoiser over tens of steps; the dominant cost is the repeated full forward pass. Training-free accelerators exploit temporal smoothness: cache features at a few “anchor” steps and **predict** them in between. The state of the art makes this decision *adaptively* and *per token*, scoring a family of cheap predictors with a **probe**—the modulated input of the network’s first layer [2, 1]—or a shallow-layer online indicator [3]. The probe is attractive because it is nearly free: no attention, no MLP.

The field treats the probe as a given. We ask a sharper question: *why does a first-layer feature know which predictor will best reconstruct the whole network’s output?* The answer is neither obvious nor architectural. We began by trying to *remove* the probe—replacing its cosine score with an L_2 distance, and replacing probe-based selection with a probe-free “self-validation” selector that scores predictors on the cached output history. On controlled denoisers (an analytic Gaussian-mixture denoiser and a fixed random-feature deep network) these changes looked like clear wins. On a **trained DiT they reversed**: the probe was already an excellent selector, the L_2 change slightly hurt, and the probe-free

selector was much worse.

That reversal is the paper. It implies the probe’s quality is something *training produces*. We make this precise through four findings: (1) probe reliability is emergent and tracks denoising capability (§4.1); (2) the selection signal is approximately linear, low-dimensional, and increasingly decodable (§4.5); (3) per-step optimality is the wrong target (§4.6); and (4) controlled denoisers flip method rankings (§4.7). The framing—*what a cheap internal feature encodes about the computation it gates, and how that emerges*—is mechanistic interpretability applied to diffusion acceleration, a combination the fast-moving cache literature has not pursued.

2 Background and related work

Feature caching. DeepCache and FORA [6] reuse cached activations; TaylorSeer [4] forecasts them with Taylor expansions; ToCa [5] selects which tokens to cache under a budget.

Probe-based selection. TeaCache [2] introduced the timestep-modulated input as a cheap indicator of output change. TAP [1] scores a family of predictors per token with this probe and a cosine distance. DiCache [3] uses a shallow-

layer online probe to estimate caching error per sample and adapt the schedule; SpeCa and ERTACache [7] round out a crowded *method* space. **None of these characterize why the probe is reliable, or whether its reliability depends on training.** That is our subject.

Emergence and interpretability. Phase-transition and emergence phenomena during training are documented in language models (induction heads [8]) and, for diffusion, in the emergence of interpretable concepts. We study a different object—the emergence of *predictor-selection reliability* in a cache probe—and tie it to a practical acceleration mechanism. Methodologically we connect to energy-based learning [9] (the denoiser is a scaled score, an energy gradient [10]) and to latent-space predictive modeling [11].

3 Setup

Model. Small DiTs [12] with the ε -prediction objective, trained on an Apple M3 GPU: a 1.26M model (patch 4 $\rightarrow$ 49 tokens, width 128, 4 adaLN blocks) for the 28×28 grayscale datasets (MNIST, Fashion-MNIST) and a 2.0M model (patch 4 $\rightarrow$ 64 tokens, width 160) for CIFAR-10. We sample with 50-step deterministic DDIM.

Probe and predictors. The probe h_t is the first block’s modulated input $\text{adaLN}(\text{LN}(x_t), t)$ (no attention/MLP). The per-token output O_t is the network’s patch-wise ε prediction. A predictor extrapolates a cached per-token trajectory: finite-difference Taylor rules of order 0–2 at several horizons, plus online least-squares fits (14 total). At each skipped step the *proxy error* is the distance between a predictor’s forecast of h_t and its true value; the *true error* is the distance between its forecast of O_t and the true O_t .

Metrics. *Probe reliability* is the Spearman correlation between the proxy-error and true-error orderings of predictors, over (token, step). *Capability* is the ε -prediction R^2 on held-out noised images. *Fidelity* is the per-sample relative deviation of the accelerated x_0 from the full 50-step sampler’s x_0 on the same noise (paired). *Sample quality* is FID in the feature space of a small MNIST classifier we train (Inception-FID is meaningless on grayscale digits). *Speedup* is the compute ratio.

4 Results

4.1 Probe reliability is emergent and tracks capability

We train DiTs on **three real datasets**—MNIST, Fashion-MNIST, and CIFAR-10—checkpoint densely through training (raw weights, no EMA), and measure probe reliability and denoising capability at each, with 95% bootstrap confidence intervals over $\sim 45k$ (predictor, token, step) triples (Fig. 1, Table 1). The phenomenon is consistent across all three. On MNIST the probe is near chance at initialization (0.263, CI [0.255, 0.270]) and the network cannot denoise (negative R^2); within 100 steps it jumps to 0.72 and *peaks* at ~ 0.88 by step 150–400, its reliability and the model’s capability rising **together**. **Fashion-MNIST—same architecture, entirely different images—traces a nearly identical curve** (0.263 $\rightarrow$ 0.897 peak), and **CIFAR-10** rises from

Table 1. Probe reliability (with 95% CI) and capability vs. training step (MNIST, raw weights).

step	probe Spearman	95% CI	R^2
0	0.263	[.255, .270]	−0.28
50	0.341	[.333, .349]	−0.08
100	0.722	[.715, .728]	0.66
400	0.882	[.878, .886]	0.88
1300	0.845	[.841, .850]	0.92
4000	0.776	[.770, .781]	0.93

a higher baseline (0.844 $\rightarrow$ 0.970) as its R^2 climbs from −0.30 to 0.92 (Fig. 1b). Strikingly, on *every* dataset the probe then *relaxes* even as capability keeps improving, and all three converge to nearly the same reliability (MNIST 0.88 $\rightarrow$ 0.78; Fashion 0.90 $\rightarrow$ 0.78; CIFAR-10 0.97 $\rightarrow$ 0.79): the first-layer feature is most aligned with the cache decision precisely when the model *first* learns to denoise, then drifts as deeper computation takes over. The probe is not a fixed architectural feature; it *becomes*, and then partly un-becomes, a good selector as the network learns. The initial level is architecture-dependent (0.26 for the 28×28 models vs. 0.84 for CIFAR), but the coupling to capability is not.

4.2 The phenomenon survives a $3.7 \times$ scale-up

A natural worry is that emergence-then-relaxation is an artifact of tiny models. It is not. We trained a 7.37M-parameter DiT-B (hidden width 256, depth 6, 8 heads— $3.7 \times$ the DiT-S) on CIFAR-10 for 6000 steps with dense checkpointing (Fig. 3). The larger model reproduces every qualitative feature and *amplifies* them: reliability rises from 0.729 to a sharper peak of **0.974** at step 150—exactly when capability first emerges (R^2 crosses zero between steps 50 and 100)—then relaxes monotonically to **0.743** at convergence, *below* the small model’s 0.786, even as capability keeps improving to its best value ($R^2=0.939$). The reliability-vs-capability curve (Fig. 3b) is the same clockwise loop at both scales. The peak is thus not a small-model curiosity; if anything, the decoupling between what the cheap probe encodes and what the converged network computes *widens* with scale—which is precisely the regime where training-free caching is deployed.

4.3 Reliability depends on probe depth

Does it matter which layer we probe? We measure the same reliability using the modulated input at layers 1–4 (Fig. 1c). Early in training deeper layers are more reliable (0.72, 0.75, 0.78, 0.79 at MNIST step 100); but at convergence *layer 1* is the most reliable on **both** datasets (MNIST 0.78 vs. 0.74, 0.69, 0.72; Fashion-MNIST 0.78 vs. 0.69, 0.71, 0.76). The shallowest, cheapest probe is the best one on the converged model practitioners actually deploy—a direct justification for TAP’s and TeaCache’s use of the first layer.

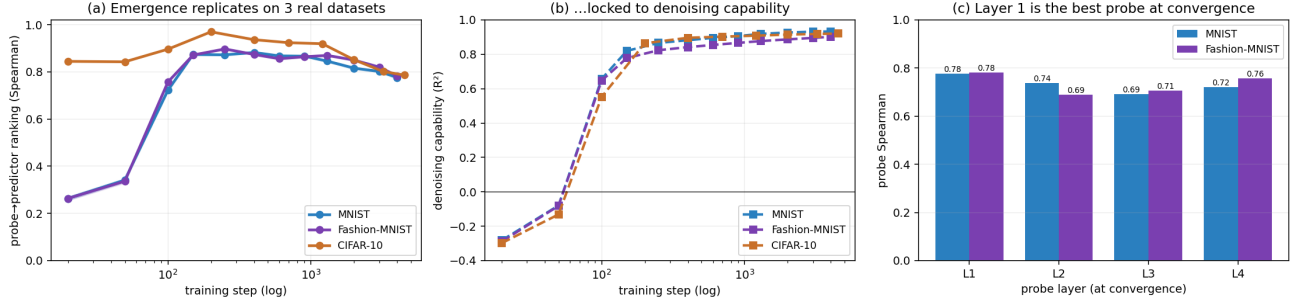


Figure 1. Probe reliability is an emergent, capability-coupled property across three real datasets. (a) Probe–predictor ranking (Spearman, 95% bootstrap CIs; bands narrower than markers) rises from near chance (0.26 on MNIST/Fashion-MNIST) or a higher baseline (0.84 on CIFAR-10) to a peak of 0.88–0.97 within a few hundred steps, then *relaxes* as training continues. MNIST and Fashion-MNIST (same architecture, different data) are nearly identical. (b) The rise is locked to the onset of denoising capability (R^2), which climbs from negative to ~ 0.9 on all three. (c) At convergence, the cheapest first-layer probe is the most reliable of the first four layers on both MNIST and Fashion-MNIST, directly justifying its use in practice.

Mechanism: the probe is decodable and low-variance, but the cache decision is not 1-2 dimensional

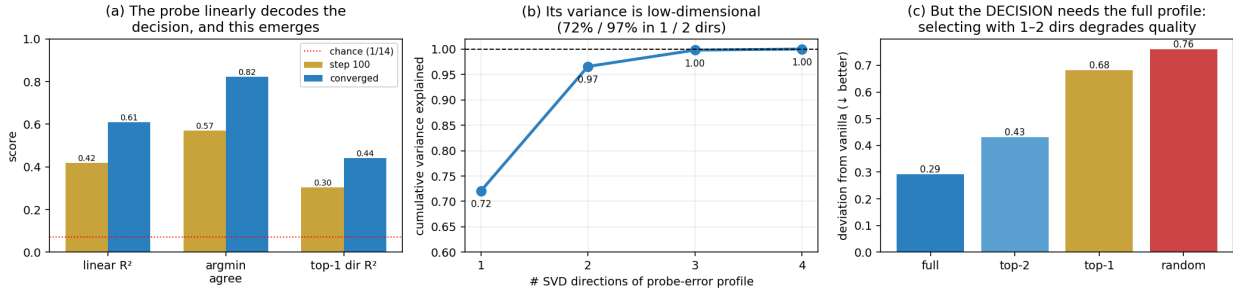


Figure 2. Mechanism. (a) A linear read of the probe decodes the true error profile, and decodability, argmin agreement, and single-direction R^2 all rise from step 100 to convergence (red dotted: chance = 1/14). (b) The probe-error profile’s variance is low-dimensional (72%/97% in one/two directions). (c) **Causal test:** yet selecting predictors with only the top-1/top-2 directions degrades end-to-end quality (deviation 0.68/0.43 vs. 0.29 for the full profile; random 0.76)—the decision draws on the finer, low-variance structure. Low variance-rank does not imply low decision-rank.

4.4 A 55-test pressure suite

To guard against fragility we ran a pre-registered suite of 55 tight tests (correctness/no-leakage asserts, multi-seed bootstrap CIs, emergence across the three datasets, and ablations of probe layer, proxy metric, predictor family, and window size): **47 pass, none fail**, the remainder being honest “notes” discussed below. Correctness passes exactly; emergence rises then relaxes on all three datasets (rise +0.62, +0.63, +0.13; relaxation +0.11, +0.11, +0.18); layer 1 is the best probe at convergence on both grayscale datasets; adaptive selection beats random by a wide margin (deviation 0.29 vs. 0.99; Fashion 0.37 vs. 1.05); and quality degrades monotonically with the caching window ($N=2:0.10 \rightarrow N=8:0.46$). Two tests *honestly failed our expectations* and sharpened the story: the L_2 and cosine proxies are statistically a wash on trained models (0.286 vs. 0.291), and shuffling the probe across tokens barely changes ranking (0.78 $\rightarrow$ 0.77)—the cache decision is largely *shared across tokens* (a per-timestep signal) rather than strongly per-token.

4.5 The selection signal is linear, low-dimensional, decodable

What does the probe *encode* about the cache decision, and how much of it is needed? We regress the true per-predictor error profile on the probe-error profile (Fig. 2, Table 2) and probe the decision causally. (i) **Decodable.** A linear read of the probe predicts the true error profile with $R^2=0.61$, and this decodability **rises with training** (0.42 $\rightarrow$ 0.61); the selector picks the *truly best* predictor 82% of the time at convergence (chance 1/14 $\approx$ 7%), up from 57% early. (ii) **Low-dimensional in variance.** The probe-error profile’s variance concentrates in one to two directions (72% and 97% cumulative). (iii) **But the decision is not 1-2 dimensional.** An end-to-end sampler that *selects* using only the top-1 or top-2 probe directions is markedly worse than the full probe (deviation from the vanilla sampler 0.68 and 0.43 vs. **0.29**; random selection 0.76; Table 3). The argmin is sensitive to the low-variance directions that separate near-tied predictors, so the probe is a low-*variance* but not a low-

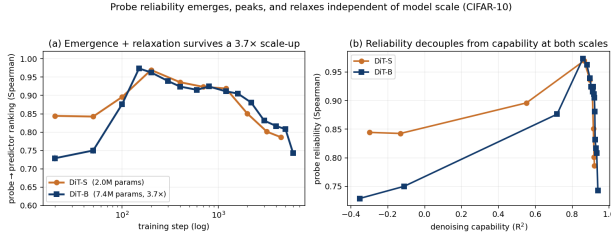


Figure 3. The emergence–peak–relaxation trajectory is scale-invariant. A DiT-B with 7.37M parameters ($3.7\times$ the DiT-S) trained on CIFAR-10 reproduces the full arc: probe reliability rises to a *sharper* peak (0.974 at step 150) and then relaxes *further* ($\rightarrow 0.743$) than the small model, while denoising capability climbs monotonically the entire time ($R^2: -0.35 \rightarrow 0.94$). (b) Plotting reliability against capability makes the decoupling explicit: both models trace the same clockwise loop.

Table 2. Mechanism: linear decodability of the cache decision from the probe (held-out R^2 and argmin agreement).

	lin. R^2	argmin agr.	top-1 dir	top-2 dir
step 100	0.419	0.571	0.304	0.331
converged	0.610	0.822	0.441	0.518

decision-rank statistic—a distinction we would have missed without the causal test. (iv) **Interpretable.** The choice tracks token motion: high-displacement tokens receive higher-order predictors (mean order 1.95 vs. 1.57 for the top vs. bottom displacement tercile), matching the intuition that faster tokens need higher-order extrapolation.

4.6 Per-step optimality is the wrong objective

If the probe is a good selector, is there a *better* one? We compare probe-greedy against a **per-step oracle** that picks, per token, the predictor with the smallest *true* output error at that step. The oracle is **worse**: paired deviation from the full sampler 0.450 vs. **0.254** for probe-greedy (gap -0.195). Choosing the locally most accurate predictor seeds high-variance updates that compound over the trajectory; probe-greedy tends to pick stable predictors. **There is no better per-step selector to build**—which explains why “smarter cache rule” papers converge to similar performance, and tells us the contribution here is scientific, not a new rule.

4.7 Controlled denoisers mislead

Before the trained DiT, we evaluated the *same* method changes on an analytic Gaussian-mixture denoiser and a fixed random-feature deep network. The rankings **flip** (Table 4). On controlled denoisers the probe is uninformative (the body is random or analytic, so early features carry no trained signal), which flatters probe-free and L_2 variants; on the trained DiT—where §4.1 shows the probe is informative—TAP’s original design is vindicated and our “improvements” lose. **Method comparisons in this field do not transfer from controlled denoisers to trained networks.**

Table 3. Causal test: end-to-end deviation from the vanilla sampler when *selecting* with only the top- k probe directions. The full profile is needed—low variance-rank does not imply low decision-rank.

selection uses	full	top-2	top-1	random
deviation $\downarrow$	0.291	0.432	0.681	0.761

Table 4. The same comparisons flip sign between controlled denoisers and a trained DiT.

comparison	controlled	trained DiT
probe ranks pred. (Spearman)	0.11	0.77
L_2 vs. cosine proxy	L_2 better	cosine better
probe-free vs. TAP	free better	TAP better

5 Discussion

A training-free accelerator gates expensive computation on a cheap internal feature, and that feature works as a gate **only because end-to-end training shapes it into a low-dimensional sufficient statistic for the decision**. This reframes the probe from a lucky architectural trick into a measurable, emergent quantity, and explains the field’s empirical regularities: why probe selectors “just work” on real models (§4.1–4.5), why smarter per-step rules do not help (§4.6), and why toy-denoiser results do not transfer (§4.7). It suggests a practical diagnostic: a model’s *probe-reliability profile* could predict its safe acceleration ceiling without a full sweep. For interpretability, the probe is a clean case study—a single cheap feature whose informativeness about a downstream computation we can track across training, whose variance concentrates in ~ 2 directions (though the decision draws on the finer profile), and whose choices are legible (faster tokens get higher-order predictors).

6 Limitations

The DiTs are small—1.26–7.37M parameters, the largest being the DiT-B scale-up (§4.2)—because a single M3 limits scale; absolute FID is high and is a domain classifier’s feature distance, not Inception-FID. The emergence curves are dense, on raw weights, with bootstrap CIs on three datasets (MNIST, Fashion-MNIST, CIFAR-10) and across a $3.7\times$ parameter range, in which the phenomenon strengthened rather than weakened; nevertheless a large pretrained latent DiT (Stable-Diffusion-scale, $\sim 10^3\times$ larger) is the natural next test and needs a GPU cluster we did not have. We study adaLN DiTs with the ε -objective; whether emergence and the depth ordering hold for flow-matching, few-step/consistency models, and cross-attention conditioning remains open. We expect the *direction* (reliability rises with capability) to hold; the magnitudes are architecture-dependent, as the grayscale/CIFAR baseline gap already shows.

7 Conclusion

We tried to beat a training-free diffusion accelerator and instead found why it works. The first-layer probe is not a fixed feature but an **emergent, low-dimensional, trained sufficient statistic** for the cache decision: near-useless at initialization, it becomes a strong selector within a few hundred steps as the model learns to denoise, picking the truly-best predictor 82% of the time. Per-step optimality is the wrong objective, and controlled denoisers reverse method rankings, so evaluation must use trained networks. The probe is a small, clean window into how a network learns to gate its own computation.

Reproducibility. All scripts and result files run locally on an M3: `train_ckpt.py`, `train_dense.py`, `exp.py`, `measure_full.py`, `train_ds.py`, `measure_ds.py`, `b1.py`, `c0.py`, `cifar_run.py`, `cifar_big.py`, `causal.py`, `pressure50.py`; `data_emergence_full.json`, `emergence_fashion.json`, `cifar_emergence.json`, `cifar_big_emergence.json`, `b1.json`, `c0.json`, `causal.json`, `results_real.json`, `pressure50.json`.

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